

YEN-LIN (ALLEN) LU

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EDUCATION

Cornell Tech, Cornell University <i>Master of Engineering in Computer Science</i>	Aug 2026 – May 2027 GPA: In Progress
New York University Tandon School of Engineering <i>Bachelor of Science in Mathematics & Computer Science; Dean's List for all academic years</i>	Sep 2023 – May 2026 GPA: 3.73/4.00
University of California, Irvine <i>Bachelor of Science in Mathematics; Dean's Honor List for all quarters</i>	Sep 2022 – Jun 2023 GPA: 3.99/4.00

TECHNICAL SKILLS

Languages: Python, C++, C, R, SQL, JavaScript, TypeScript, Java, Swift, HTML/CSS, Node.js, LaTeX, MATLAB
Frameworks & Libraries: PyTorch, FastAPI, Express.js, Flask, React, Vite, SwiftUI, ActivityKit, scikit-learn, XGBoost, SHAP, Pandas, NumPy, Jest, Vitest, XCTest, SQLAlchemy, Mongoose, Tailwind CSS, Bootstrap
Tools & Databases: PostgreSQL, TimescaleDB, Redis, MongoDB, SQLite, MySQL, Docker, Nginx, WebSockets, Git/GitHub, Postman, AWS S3

EXPERIENCE

Research Assistant <i>NYU Cognition and Computation Lab</i>	Oct 2025 – May 2026 New York, NY
- Built a dual LLM-Bayesian learning system using PyTorch, Transformers, and LLMs to translate natural language into symbolic meaning functions for explainable AI - Designed in-context learning pipelines and ran benchmark experiments comparing LLM-generated vs. hand-coded functions for probabilistic inference	
Student Product Support Analyst <i>New York University</i>	Oct 2024 – May 2026 New York, NY
- Tested and triaged Brightspace platform updates pre-deployment, ensuring stability and QA across 80,000+ users - Authored technical documentation and delivered 50+ live training sessions, reducing recurring support tickets	
Software Development Engineer Intern <i>Athleo.ai</i>	Jun 2024 – Aug 2024 Boston, MA (Remote)
- Engineered a responsive website using HTML, CSS, and JavaScript, driving 1,500+ daily views - Resolved 20+ client-reported bugs through structured QA testing and debugging, improving product reliability - Produced technical onboarding videos viewed 30,000+ times, streamlining user onboarding and documentation	

PROJECTS

QuantStream <i>TypeScript, Python, PostgreSQL, TimescaleDB, Redis, PyTorch, Docker</i>	Sep 2025 – Jun 2026
- Engineered an algorithmic trading platform with async pipelines for ML signal generation and execution - Reduced market data query latency using Redis caching and TimescaleDB for time-series storage - Containerized the system with Docker and Nginx, validating core functionality through 200+ unit/integration tests	
DBook <i>React, Vite, Node.js, Express.js, MongoDB, Mongoose, Passport.js, Groq API</i>	Dec 2025 – May 2026
- Designed a digital bookshelf utilizing a MongoDB metadata cache layer to optimize external API latency - Orchestrated asynchronous recommendation pipelines via Groq API (Llama 3.3 70B) with user analytics charts - Devised session authentication via Passport.js and custom RBAC middleware to secure admin and user routes	
EcoBite <i>React, Python, SQLite, SQLAlchemy, Mapbox GL, OpenAI API, Tailwind CSS</i>	Dec 2025 – May 2026
- Synthesized multi-source NYC Open Data registries via an algorithmic pipeline calculating green score rankings - Constructed an OpenAI GPT-4o-mini Vision computer vision pipeline to extract unstructured menu data & metrics - Formulated a responsive FastAPI backend with Mapbox GL to orchestrate interactive GeoJSON polygon mapping	
IAL Trading Platform <i>JavaScript, Node.js, Express.js, MongoDB, Bootstrap, Jest</i>	Sep 2025 – Dec 2025
- Created a full-stack trading application featuring fractional share orders, portfolio analytics, and P&L tracking - Architected an Express.js API with a MongoDB backend, bcrypt authentication, and a Bootstrap UI - Wrote 33+ Jest unit tests achieving 100% code coverage for all trading validation and financial calculations	

LEADERSHIP & INVOLVEMENT

Taiwan Armed Forces – Mandatory Military Service (Summer 2025, Summer 2026) | **Applied Math Club** – Co-Founder, Secretary, Treasurer | **NYU Theta Tau Professional Engineering Fraternity** – Webmaster, Treasurer | **NYU VIP RoboMaster Team** – Software Team Member